Rewrite by completing the square.

original equation: x^2+x+1

Note: complete the square means:

find constants b, c such that

$$x^2+x+1 = (x+b)^2+c$$

$$\Leftrightarrow x^2+2xb+b^2+c$$

$$\Leftrightarrow 2b=1 \text{ and } b^2+c=1$$

$$\Rightarrow b=\frac{1}{2} \text{ and } \left(\frac{1}{2}\right)^2+c=1$$

$$\Rightarrow b=\frac{1}{2} \text{ and } c=1-\frac{1}{4}=\frac{3}{4}$$

$$\Rightarrow x^2+x+1 = \left(x+\frac{1}{2}\right)^2+\frac{3}{4}$$

$$\therefore \left(x+\frac{1}{2}\right)^2+\frac{3}{4}$$